HTML Notes: Chapter 6 – Tables

Table structure, headings, spanning cells, long tables

Basics

- Some information is best shown in a grid or table, e.g. sports results, stock reports, train timetables.
- Think of a grid of **rows** and **columns**, a bit like a spreadsheet.
- This chapter: use the four key elements for tables (`<table>`, `<tr>`, `<td>`, `<th>`), represent complex data, and add captions.

Basic table structure `<table>` `<tr>` `<td>`

- `<table>` creates the table. Contents are written out **row by row**.
- `<tr>` = table row. Opening `<tr>` starts a row, closing `</tr>` ends it.
- `<td>` = table data (one cell). One `<td>` for each cell in the row, each closed with `</td>`.
- Some browsers draw lines around the table and/or cells automatically. Borders are controlled with CSS.

HTML

```
<table>
  <tr>
    <td>15</td>
    <td>15</td>
    <td>30</td>
  </tr>
  <tr>
    <td>45</td>
    <td>60</td>
    <td>45</td>
  </tr>
  <tr>
    <td>60</td>
    <td>90</td>
    <td>90</td>
  </tr>
</table>
```

RESULT

15	15	30
45	60	45
60	90	90

(Borders drawn here only to show the cells.)

Table headings `<th>`

- `<th>` = table heading. Used like `<td>` but it is the heading for a **column or a row**.
- Even an empty cell needs a `<td>` or `<th>`, or the table won't render correctly (the first cell in the first row here is empty).
- Using `<th>` helps screen reader users, helps search engines index your pages, and gives you more control over looks with CSS.
- `scope` on `<th>` says what it heads: `scope="col"` = heading for a column, `scope="row"` = heading for a row.
- Browsers usually show `<th>` content bold and centred in the cell.

HTML

```
<table>
  <tr>
    <th></th>
    <th scope="col">Saturday</th>
    <th scope="col">Sunday</th>
  </tr>
  <tr>
    <th scope="row">Tickets sold:</th>
    <td>120</td>
    <td>135</td>
  </tr>
  <tr>
    <th scope="row">Total sales:</th>
    <td>$600</td>
    <td>$675</td>
  </tr>
</table>
```

RESULT

	Saturday	**Sunday**
Tickets sold:	120	135
Total sales:	$600	$675

Spanning columns `colspan`

- Sometimes an entry needs to stretch across more than one column.
- `colspan` goes on a `<th>` or `<td>` and says how many columns the cell runs across.
- Here: 5 columns. The first holds the row heading (the day), the other four are one-hour slots.
- 'Geography' has `colspan="2"` (two columns). 'Gym' has `colspan="3"` (three columns).
- So the 2nd and 3rd rows have **fewer `<td>` elements** than there are columns: the cells that would sit under the wider cell are left out of the code.

HTML

```
<table>
  <tr>
    <th></th>
    <th>9am</th>
    <th>10am</th>
    <th>11am</th>
    <th>12am</th>
  </tr>
  <tr>
    <th>Monday</th>
    <td colspan="2">Geography</td>
    <td>Math</td>
    <td>Art</td>
  </tr>
  <tr>
    <th>Tuesday</th>
    <td colspan="3">Gym</td>
    <td>Home Ec</td>
  </tr>
</table>
```

RESULT

	9am	10am	11am	12am
Monday	Geography		Math	Art
Tuesday	Gym			Home Ec

Spanning rows `rowspan`

- Entries can also stretch **down** across more than one row.
- `rowspan` goes on a `<th>` or `<td>` and says how many rows the cell spans down the table.
- Here ABC shows a movie from 6pm to 8pm (`rowspan="2"`), while BBC and CNN each show two one-hour programmes.
- The last `<tr>` has only 3 elements even though there are 4 columns, because the movie in the row above stretches down and takes over the cell below.

HTML

```
<table>
  <tr>
    <th></th>
    <th>ABC</th>
    <th>BBC</th>
    <th>CNN</th>
  </tr>
  <tr>
    <th>6pm - 7pm</th>
    <td rowspan="2">Movie</td>
    <td>Comedy</td>
    <td>News</td>
  </tr>
  <tr>
    <th>7pm - 8pm</th>
    <td>Sport</td>
    <td>Current Affairs</td>
  </tr>
</table>
```

RESULT

	ABC	BBC	CNN
6pm - 7pm	Movie	Comedy	News
7pm - 8pm		Sport	Current Affairs

Long tables `<thead>` `<tbody>` `<tfoot>`

- Three elements separate the main content of a table from the first and last rows (which can hold different content).
- They help screen reader users and let you style these sections differently from the rest of the table with CSS.
- `<thead>` = the headings of the table.
- `<tbody>` = the body of the table.
- `<tfoot>` = the footer of the table.
- By default browsers rarely treat these any differently, but designers often use CSS to change how they look.

HTML

```
<table>
  <thead>
    <tr>
      <th>Date</th>
      <th>Income</th>
      <th>Expenditure</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th>1st January</th>
      <td>250</td>
      <td>36</td>
    </tr>
    <tr>
      <th>2nd January</th>
      <td>285</td>
      <td>48</td>
    </tr>
    <!-- additional rows -->
    <tr>
      <th>31st January</th>
      <td>129</td>
      <td>64</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td></td>
      <td>7824</td>
      <td>1241</td>
    </tr>
  </tfoot>
</table>
```

Long tables: the full result

- This is what the long table looks like with all 31 rows: **thead** at the top, **tbody** with the daily rows, **tfoot** with the totals at the bottom.
- Some HTML editors in content management systems have tools to help draw tables. If the first row of your table only contains `<th>` elements, the editor may insert a `<thead>` automatically.
- Part of the reason for separate `<thead>` and `<tfoot>`: if a table is taller than the screen (or longer than one printed page), the browser can keep the header and footer visible while the body scrolls, so users can see which column the data is in.
- **But** this is not implemented by default in any current browser.
- The book added CSS here so you can see the thead and tfoot treated differently from the other rows (CSS covered later).

RESULT

Date	Income	Expenditure
1st January	250	36
2nd January	285	48
3rd January	260	42
4th January	290	38
5th January	310	115
6th January	168	14
7th January	226	20
8th January	253	37
9th January	294	33
10th January	216	46
11th January	244	29
12th January	297	32
13th January	328	86
14th January	215	38
15th January	254	30
16th January	256	27
17th January	311	68
18th January	212	39
19th January	234	36
20th January	221	43
21st January	259	38
22nd January	246	31
23rd January	248	17
24th January	229	45
25th January	263	34
26th January	258	41
27th January	283	22
28th January	256	30
29th January	278	47
30th January	251	15
31st January	129	64
	7824	1241

Old code: width & spacing

- These are **outdated** attributes. Don't use them on new sites, but you may see them in older code. All have been replaced by CSS.
- `width`: used on the opening `<table>` tag for the width of the table, and on some `<th>` / `<td>` tags for the width of individual cells. The value is in pixels.
- Columns must form a straight line, so `width` is often only on the **first row** (later rows follow that setting).
- `cellpadding`: space **inside** each cell.
- `cellspacing`: space **between** the cells. Both values are in pixels.
- Use CSS to control the width or spacing of tables and cells.

HTML

```
<table width="400" cellpadding="10"
  cellspacing="5">
  <tr>
    <th width="150"></th>
    <th>Withdrawn</th>
    <th>Credit</th>
    <th width="150">Balance</th>
  </tr>
  <tr>
    <th>January</th>
    <td>250.00</td>
    <td>660.50</td>
    <td>410.50</td>
  </tr>
  <tr>
    <th>February</th>
    <td>135.55</td>
    <td>895.20</td>
    <td>1170.15</td>
  </tr>
</table>
```

RESULT

	Withdrawn	Credit	Balance
January	250.00	660.50	410.50
February	135.55	895.20	1170.15

(Padding inside cells made bigger here to show cellpadding. The book added CSS to show the cell widths.)

Old code: border & background

- Also outdated. You may see them in older websites. Use CSS for new sites.
- `border`: used on both `<table>` and `<td>` to set the border width in pixels.
- `bgcolor`: sets the background colour of the whole table or individual cells. The value is usually a hex code (e.g. #efefef, #cccccc).
- This example uses only the HTML `border` and `bgcolor` attributes, no CSS.

HTML

```
<table border="2" bgcolor="#efefef">
  <tr>
    <th width="150"></th>
    <th>Withdrawn</th>
    <th>Credit</th>
    <th width="150"
      bgcolor="#cccccc">Balance</th>
  </tr>
  <tr>
    <th>January</th>
    <td>250.00</td>
    <td>660.50</td>
    <td bgcolor="#cccccc">410.50</td>
  </tr>
  <tr>
    <th>February</th>
    <td>135.55</td>
    <td>895.20</td>
    <td bgcolor="#cccccc">1170.15</td>
  </tr>
</table>
```

RESULT

	Withdrawn	Credit	Balance
January	250.00	660.50	410.50
February	135.55	895.20	1170.15

(Grey fills show the bgcolor values.)

Quick cheat sheet

Tag / attribute	Use
`<table>`	Creates the table
`<tr>`	Table row
`<td>`	Table data (a cell)
`<th>`	Table heading (column or row)
`scope="col" / "row"`	Says if a heads a column or a row
`colspan="n"`	Cell stretches across n columns
`rowspan="n"`	Cell stretches down n rows
`<thead>`	Heading rows
`<tbody>`	Main body rows
`<tfoot>`	Footer rows
`width (old)`	Table / cell width in pixels, use CSS
`cellpadding (old)`	Space inside cells, use CSS
`cellspacing (old)`	Space between cells, use CSS
`border (old)`	Border width in pixels, use CSS
`bgcolor (old)`	Background colour (hex), use CSS